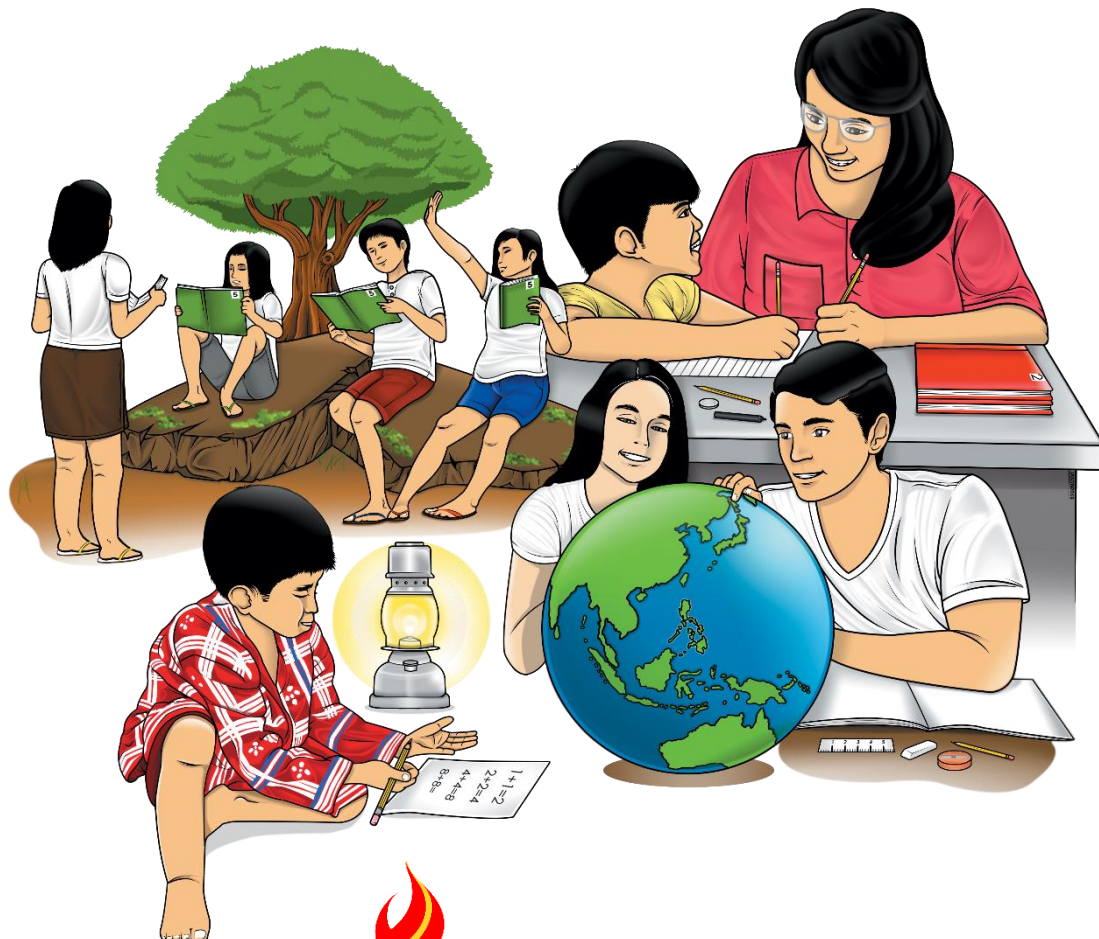


Science

Quarter 2 – Module 3:

Explain How the Different Organ System Works Together



Science – Grade 6
Alternative Delivery Mode
Quarter 2 – Module 3: Explain How the Different Organ System Works Together
First Edition, 2019

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Science

Quarter 2 – Module 3: Explain How the Different Organ System Works Together

Introductory Message

This Self-Learning Module (SLM) is prepared so that you, our dear learners, can continue your studies and learn while at home. Activities, questions, directions, exercises, and discussions are carefully stated for you to understand each lesson.

Each SLM is composed of different parts. Each part shall guide you step-by-step as you discover and understand the lesson prepared for you.

Pre-tests are provided to measure your prior knowledge on lessons in each SLM. This will tell you if you need to proceed on completing this module or if you need to ask your facilitator or your teacher's assistance for better understanding of the lesson. At the end of each module, you need to answer the post-test to self-check your learning. Answer keys are provided for each activity and test. We trust that you will be honest in using these.

In addition to the material in the main text, Notes to the Teacher are also provided to our facilitators and parents for strategies and reminders on how they can best help you on your home-based learning.

Please use this module with care. Do not put unnecessary marks on any part of this SLM. Use a separate sheet of paper in answering the exercises and tests. And read the instructions carefully before performing each task.

If you have any questions in using this SLM or any difficulty in answering the tasks in this module, do not hesitate to consult your teacher or facilitator.

Thank you.



What I Need to Know

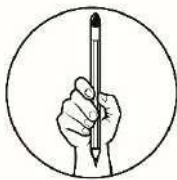
This module was designed and written with you in mind. It is here to help you master how the Major Organs of the Nervous System work with Musculo- Skeletal, Digestive, Respiratory and Circulatory System. The scope of this module is used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

This module is about:

- Explaining how the different organ systems work together

After going through this module, you are expected to:

- Explain how the major organ systems work together



What I Know

A. Directions: Choose the letter of the correct answer. Write your answer in your Science journal.

1. Which of the following is the function of integumentary system?
 - a. It serves as a body covering.
 - b. It serves as the pathway of the blood.
 - c. It serves as a sense of touch.
 - d. both a and c
2. How does musculo-skeletal system work together?
 - a. Muscles are attached to the bones and produce movement by contracting and relaxing.
 - b. Muscles are attached to the skin and produce movement by the help of the epidermis.
 - c. Muscles are attached to the ribs and spinal column; thus, they produce movement.
 - d. Muscles are attached to the cartilage, thus producing movement.

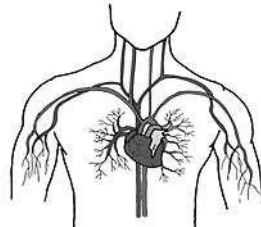
3. The following are the works of circulatory system **except**_____.
- carrying nutrients to the cells
 - carrying oxygen to the different parts of the body
 - carrying message to the brain
 - carrying waste product of the cells
4. How does circulatory system work together with respiratory system?
- Circulatory system breaks down food to be absorbed by the respiratory system as it provides oxygen to the cells.
 - Circulatory system carries nutrients to the respiratory system as it provides oxygen to the circulatory system.
 - Circulatory system carries message to the respiratory system as it provides oxygen to the circulatory system.
 - Circulatory system carries oxygen to the cells of respiratory system as it provides nutrients to the circulatory system.
5. How does digestive system work together with respiratory and circulatory systems?
- Digestive system digests food to be carried out by the circulatory system to the respiratory system as it provides oxygen to the digestive system.
 - Digestive system digests food to be carried out by the respiratory system to the circulatory system as it provides oxygen to the digestive system.
 - Digestive system carries nutrients to the circulatory system, while circulatory system digests food for the digestive system as it provides oxygen.
 - Digestive system provides food and oxygen to the circulatory and respiratory system.

B. Directions: Match the body system in Column A with its proper function in Column B. Write your answer in your Science journal.

A

B

6.



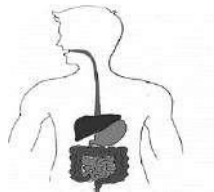
a. provides support and movement

7.



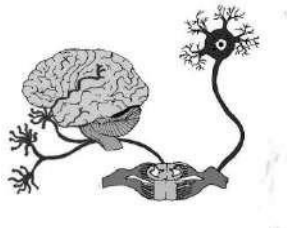
b. breaks down food into smaller substances

8.



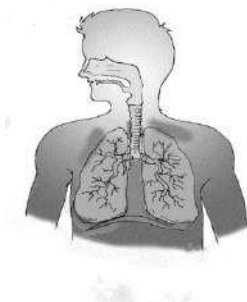
c. helps in breathing in oxygen and breathing out carbon dioxide

9.



d. carries nutrients to the cells and carries out the waste product of the cells

10.



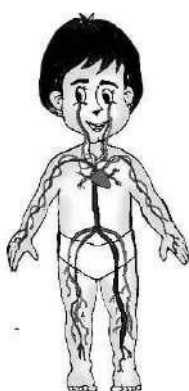
e. serves as control unit of the body

6. What is the main organ of the respiratory system?
 - a. lungs
 - b. bronchi
 - c. pharynx
 - d. nostrils
7. Which of the following is a muscular organ that pumps blood throughout the body?
 - a. atria
 - b. ventricles
 - c. septum
 - d. heart
8. The__is the outer layer of the skin where hair is located.
 - a. Dermis
 - b. hair follicles
 - c. epidermis
 - d. fatty tissues

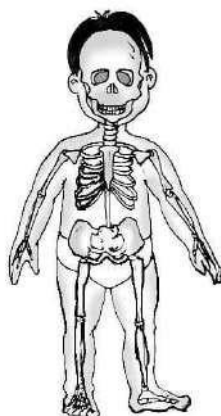


What's New

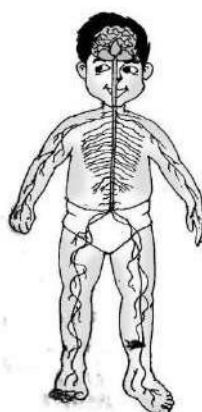
Directions: Based on the picture below, complete the table by supplying the columns with correct answers. Write your answer in your Science journal.



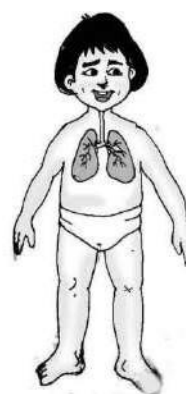
Circulatory System



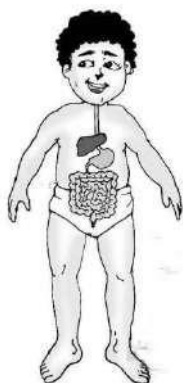
Skeletal System



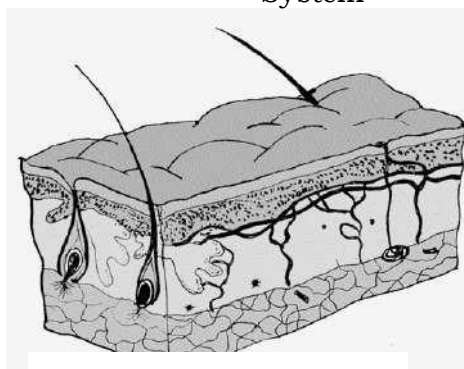
Nervous System



Respiratory System



Digestive System



Integumentary System

Body System	Organs Involved	Functions of the Body System
Integumentary System		
Muscular System		
Skeletal System		
Digestive System		
Respiratory System		
Circulatory System		
Nervous System		



What Is It

Different major organs of the body worked together to keep our body alive and produce different movements with specific functions and coordination yielding desired result. The following are the body systems and their specific functions:

1.

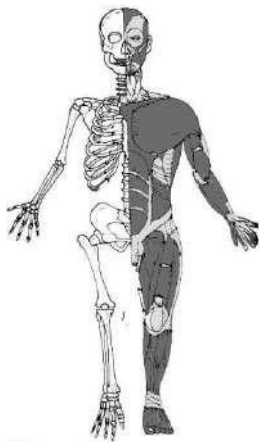


Figure 1: Musculo-skeletal system

Musculo-skeletal System

The musculo-skeletal system provides support to the body and enables the body to produce movement. Bones in the body (skeletal system) muscles (muscular system), the cartilage, tendons, ligaments, joints, and other connective tissue that supports and binds tissues and organs together comprise the musculo-skeletal system.

Muscular System – all muscles in the body make up the skeletal system which allows us to move. These consist of the three types of muscles, the cardiac muscles that control the heart, smooth muscles that control involuntary movement of the digestive tract, blood vessels,

and bladder, and the skeletal muscles which are voluntary muscles that control the movement of bones.

Skeletal System – all the bones in the body make up the skeletal system which protects internal organs and supports the body. It also produces millions of blood cells needed by the body.

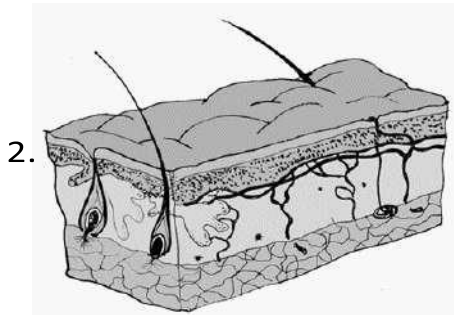


Figure 2: Integumentary system

Integumentary System

The integumentary system serves as body covering that provides protection to our body against microorganisms, temperature, and acts as receptor for touch.

The skin is composed of tough skin cells and tough protein called keratin that protects tissues, organs, and other structures underneath the skin against physical damage from minor cuts.

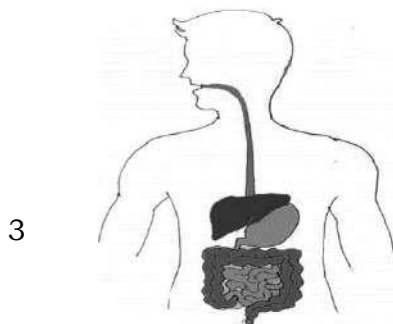


Figure 3: Digestive system

Digestive System

Digestive system works in digesting food into substances that can be absorbed by the body. Digestion begins in the mouth by breaking food by means of cutting, tearing, and grinding known as mechanical digestion.

The breakdown of food into smaller molecules by enzymes found in the digestive system is known as chemical digestion. Final digestion of food and absorption of nutrients takes place in the small intestine.

4.

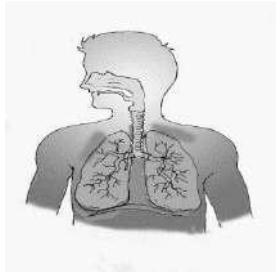


Figure 4: Respiratory system

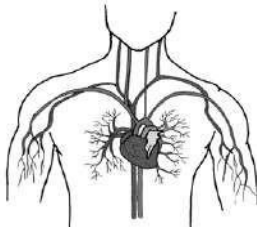
Respiratory System

Respiratory system is responsible for the exchange of gases such as carbon dioxide to oxygen in the bloodstream.

Respiratory process starts from the nostrils, pharynx, to the trachea, into the bronchi and to the lungs.

Figure 5: Circulatory system

5



Circulatory System

Circulatory system is a transport system of the body responsible for the distribution of nutrients to the cells and carries away the waste product of the cells. The heart is the vital organ in the circulatory system that pumps blood to all body parts. The blood vessels are the railway station of the blood which enables blood to reach different parts of the body. Blood is another component of the circulatory system as it circulates to all parts of the body and it transports oxygen from the lungs to the cells and carries the waste materials from the cells.

5.

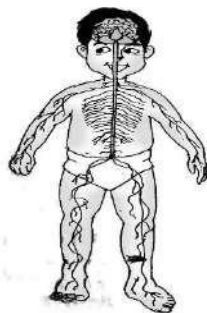


Figure 6: Nervous system

Nervous System

Nervous system is the control unit of the body; thus, it regulates and coordinates body processes. It has two major divisions: the central nervous system (CNS) composed of brain and spinal cord and the

peripheral nervous system (PNS) consists mainly of nerves that connect the CNS with the rest of the body.

Its major parts are the brain, spinal cord, and neurons or nerve cells.

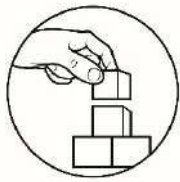
Brain is the control unit of the body as it interprets messages, makes judgments, interprets sensations, and t h i n k s creatively. The spinal cord is a cordlike material encased in a spinal column that links between the brain and the PNS. It is also responsible for a reflex action or an instantaneous movement that protects the body from harm. The neurons or nerve cells are the working unit of the nervous system as they transmit information to the nerve cells, muscles or gland cells.

Below is the information of the interactions of the **Nervous System** on the different body systems.

Table 1

System	Functions	Interactions with the Nervous System
Skeletal	1. It supports the body/framework of the body. 2. It protects the internal organs of the body. 3. The skeletal system produces blood cells. 4. It stores minerals like calcium and phosphorous.	1. The skull protects the brain from injuries. 2. The brain controls the position of the bones in the body. 3. The spinal column protects the spinal cord from injury. 4. Bones provide calcium essential for the function of the nervous system.
Muscular	1. Different types of muscles control voluntary and involuntary movement of the body.	1. The brain controls voluntary movement.

	2. Muscles produce heat to maintain body temperature. 3. Muscles move food in the digestive tract. 4. It contracts heart.	2. Muscles in the body provide information to the brain. 3. The nervous system regulates the speed at which food moves through the digestive tract.
Integumentary	1. The skin is the body covering for protection, regulates body temperature and acts as receptor for touch.	1. The receptor cells in the skin send message to the brain. 2. Brain controls the sweat glands of the body. 3. The nerves underneath the skin control muscles connected to the hair follicles
Digestive	1. The digestive system stores and digests food into simpler form. 2. It provides food to the body. 3. It eliminates waste of the body.	1. The digestive system provides food for the brain 2. Brain controls eating behavior/stomach contraction. 3. The brain controls muscles for eating and elimination. 4. The digestive system sends sensory information to the brain.
Respiratory	1. The respiratory system supplies oxygen to the blood and removes carbon dioxide.	1. The brain monitors respiratory process 2. The brain controls respiratory rate.
Circulatory	1. The circulatory system carries nutrients to the cells and carries away waste product of the cells. 2. The heart pumps blood to all parts of the body.	1. The brain controls blood and heartbeat activities. 2. The brain regulates heart rate and blood pressure.

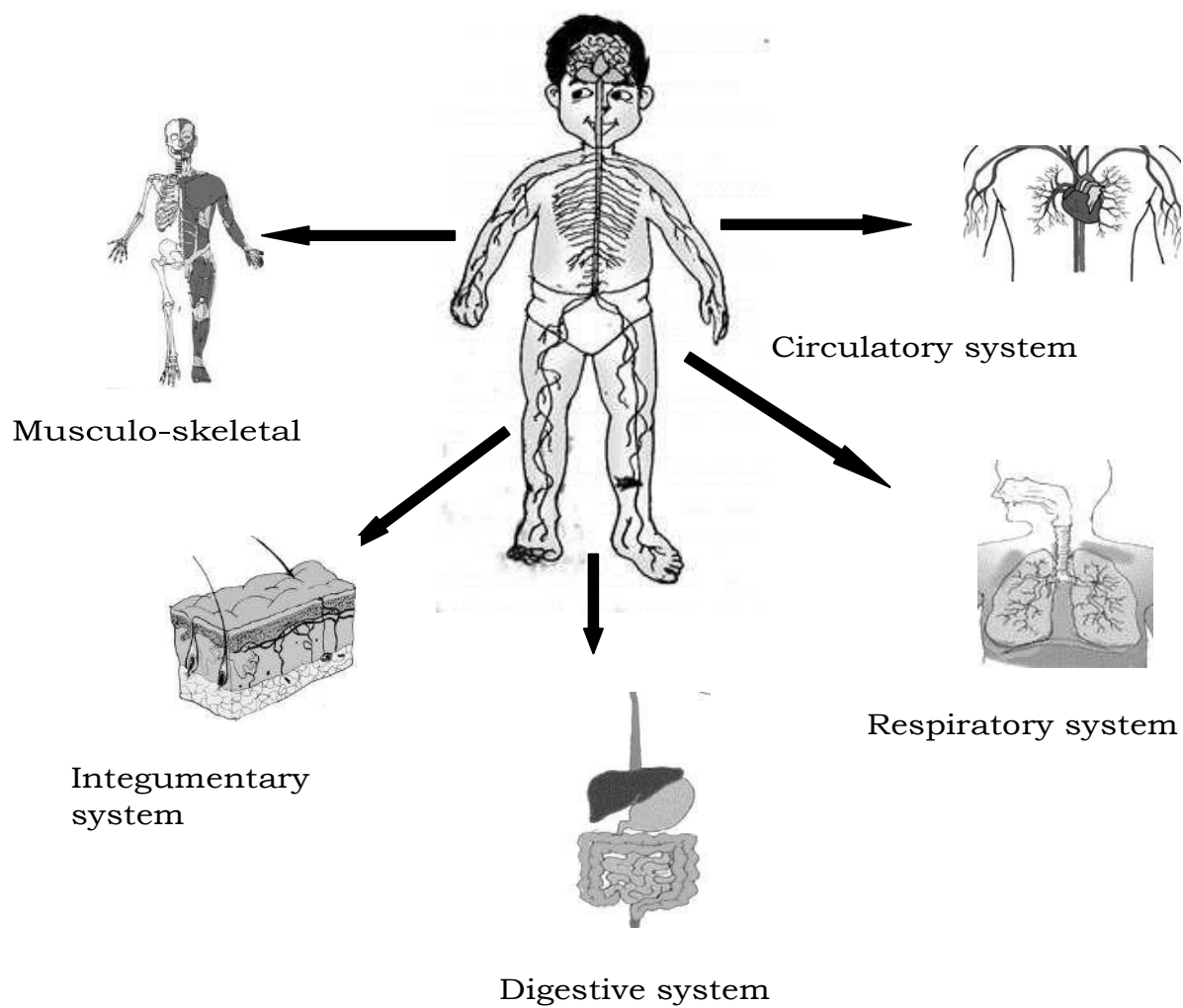


What's More

**Direction: Based on Table 1, complete the data below.
Write the answers in your journal.**

1. Describe how nervous system worked with integumentary system.
Receptor cells of the _____
_____ brainBrain controls _____
_____.
.
2. How does nervous system work with musculo-skeletal system? Skull protects_____.
The brain controls the _____
_____ and _____
movement.
3. How does nervous system work with the circulatory system? Circulatory system carries_____ and _____
_____ product of the cells.
Brain controls_____ and _____
_____ activities.
4. How does nervous system work with respiratory system? Respiratory system supplies_____ to the brain.
5. Based on the illustration below, explain how the different organ systems work together.

Nervous System





What I Have Learned

Directions: Complete the sentences by writing the correct answers using your journal.

I learned that...

Nervous system works with the following body systems such as:

Skeletal system by controlling the _____

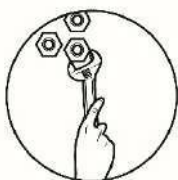
Muscular system by controlling _____

Integumentary system by controlling _____

Digestive system by controlling _____

Respiratory system by monitoring _____

Circulatory system by controlling _____ and _____.



What I Can Do

Study the chart below. Check the YES column showing healthful habit that promotes proper functioning of the organ system and check the NO column if not.

HEALTHFUL HABITS	YES	NO
1. Eat balanced diet.		
2. Have enough rest every day.		
3. Playing video games until midnight.		
4. Take a bath every day.		
5. Sunbathing from 10 – 12 noon.		
6. Sleep at least 6 hours a day.		
7. Exercise regularly.		
8. Eat too much fatty foods.		
9. Have proper ventilation and sunshine in you room.		
10. Have medical check-up regularly.		

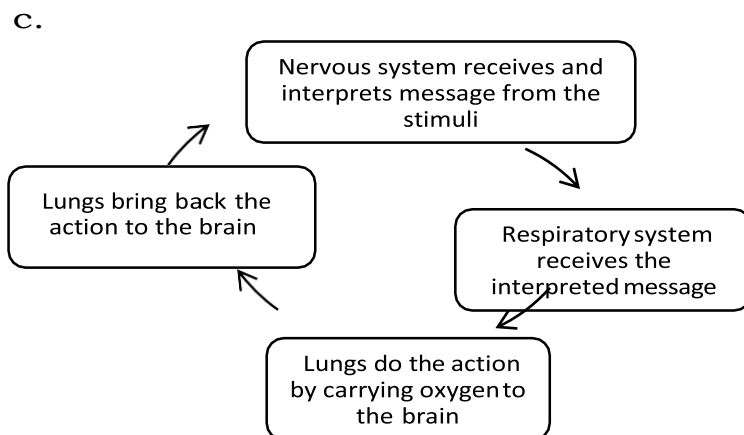
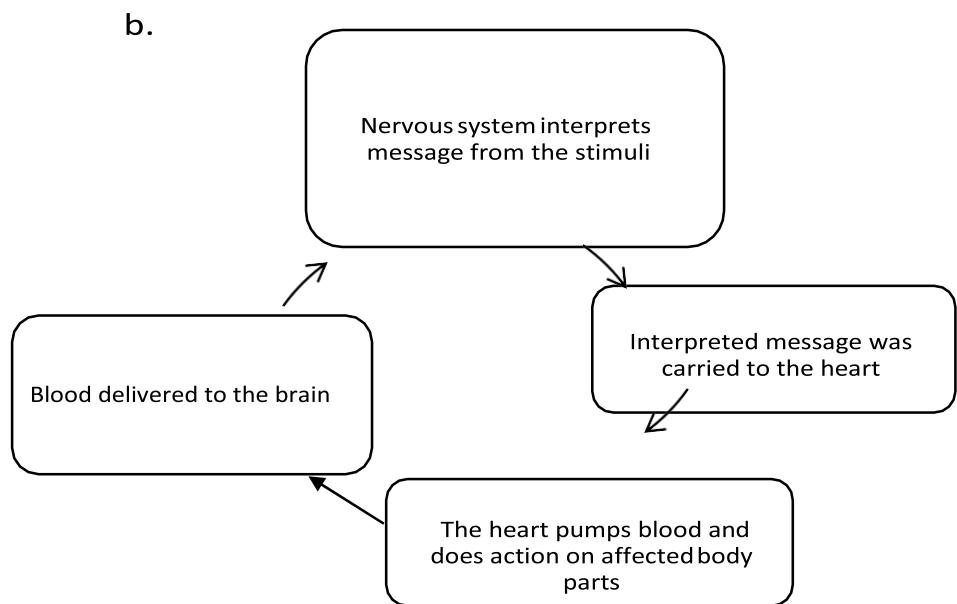
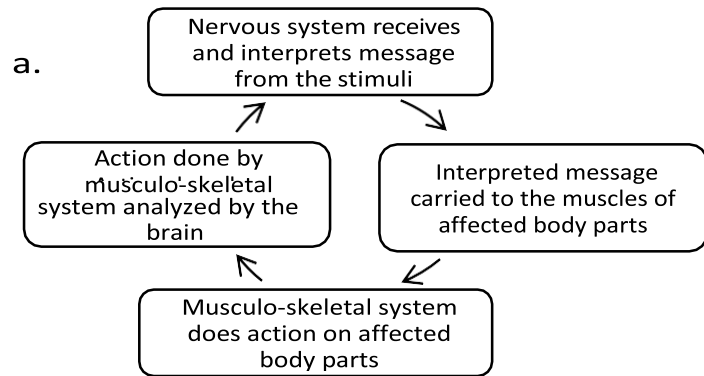


Assessment

Directions: Choose the correct answer in each number. Use a separate sheet for your answers.

1. How does musculo-skeletal system work with the integumentary system?
 - a. Musculo-skeletal system gives support to the body and helps it to move while integumentary system gives protection to the musculo-skeletal by providing a protective covering.
 - b. Musculo-skeletal system gives oxygen to the integumentary system while integumentary system provides food to the musculo-skeletal system.
 - c. Musculo- skeletal system pumps blood to integumentary system while integumentary system gives support to the musculo-skeletal system.
 - d. Musculo-skeletal system excretes sweat, thus providing pathway to the integumentary system in absorbing oxygen.
2. How does respiratory system work with circulatory system?
 - a. Respiratory system provides nutrients to the circulatory system, thus circulatory system provides oxygen to the respiratory system.
 - b. Respiratory system serves as pathway of blood carrying nutrients to the circulatory system, thus circulatory system pumps blood to the respiratory system.
 - c. Respiratory system provides oxygen to the circulator system, thus circulatory system carries blood with nutrients needed by the respiratory system.
 - d. Respiratory system carries carbon dioxide to the circulatory system, thus circulatory system provides oxygen needed by the respiratory system.

3. Analyze how the nervous system does work with musculo- skeletal system.



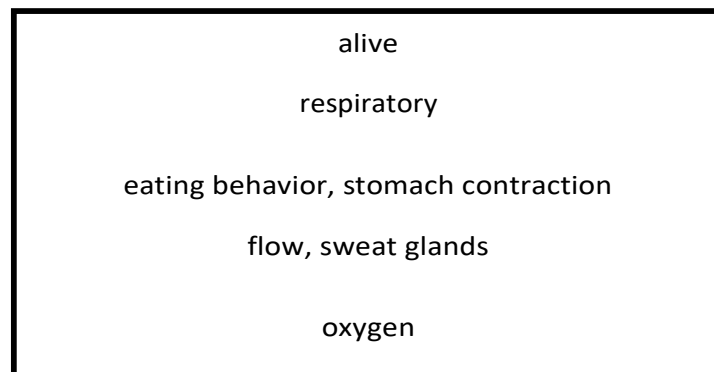
4. How does digestive system work with circulatory system?

- a. Digestive system provides oxygen to the circulatory system and circulatory system provides nutrients to the digestive system.
- b. Digestive system provides support to the circulatory system and circulatory system pumps blood to the digestive system.
- c. Digestive system digests food into simpler form which will be needed by the circulatory system and the circulatory system distributes nutrients to the body parts.
- d. Digestive system excretes waste product produced by the circulatory system and the circulatory system interprets message coming from the digestive system.

5. Which of the following is the work of circulatory system?

- a. Gives support to the body
- b. Interprets all messages
- c. Responsible for the exchange of oxygen and carbon dioxide
- d. Carries nutrients to the cells and carries away the waste product of the cells

Directions: Complete each sentence with the correct word found inside the box.



Nervous system does work with integumentary system by controlling blood _____ and _____ (6). It also works with digestive system by controlling _____ and _____ (7)

The respiratory system works with the nervous system by supplying _____ (8) to the brain, while nervous system monitors _____ (9) process. Major organs of the body work together to keep our body _____ (10)



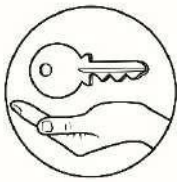
Additional Activities

Directions: Complete the chart with healthy habits that promote proper functioning of the organ systems.

Body System	Healthy Habits that Promote Proper Functioning of the Organ Systems
Integumentary System	
Musculo-skeletal System	
Digestive System	
Respiratory System	
Circulatory System	
Nervous System	

RUBRICS in Rating Additional Activity

POINTS	INDICATORS
5	Give correct answers in the 5 organ systems
4	Give correct answers in the 4 organ systems
3	Give correct answers in the 3 organ systems
2	Give correct answers in the 2 organ systems
1	Give correct answer at least in 1 organ system



Answer Key

What I Can Do 1. YES 2. YES 3. NO 4. YES 5. NO 6. NO 7. YES 8. NO 9. YES 10. YES Assessment 1. A 2. C 3. A 4. C 5. D 6. flow, sweat glands 7. eating behavior, stomach contraction 8. oxygen 9. respiratory 10. alive Additional Activities Answer may vary	What's New 1. skin-body covering/regulates body temperature 2. muscles-for movement 3. bones-supports the body 4. stomach, liver, intestines-digests food 5. lungs-exchange of oxygen and carbon dioxide 6. heart, blood, blood vessels- carries nutrients to the cells and carries away waste product of the cells 7. brain, spinal cord, nerves-controls body activities What's More 1. -skin sends message to the brain -Sweat gland of the body 2. -brain -position of the bones -movement of muscles 3. nutrients, carries away waste -blood pressure, heartbeat 4. oxygen 5. answers may vary	Lesson 1: What I Know: 1. D 2. A 3. C 4. B 5. A 6. D 7. A 8. B 9. E 10. C What's In 1. D 2. A 3. C 4. B 5. B 6. A 7. D 8. C
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